SafeRoute

"Navigate Smart. Travel Safe."

Smart Safety-Aware & Navigation System

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SLIDE 1

SafeRoute

INTRODUCTION

What is SafeRoute?

SafeRoute is a navigation system that helps people choose safer routes to their destination.

Unlike normal map applications that mainly find the fastest route, **SafeRoute** also checks safety conditions.

It considers:

- Route distance and travel time
- Street lighting
- Reported incidents
- Time of travel
- Nearby hospitals, police stations and pharmacies



In simple words:

SafeRoute helps you reach your destination through a safer route, not just a faster route.

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SLIDE 2

PROBLEM IDENTIFICATION

What is the Problem?

When we use normal navigation apps, they usually suggest the **shortest** or **fastest route.**

But the fastest route may sometimes pass through:

- Dark or poorly lit roads
- Isolated areas
- Areas with reported incidents
- Potentially risky locations
- Areas far from emergency facilities



Main Problem

A route that is faster may not always be safer.

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PROBLEM JUSTIFICATION

Why Do We Need SafeRoute?

Imagine two routes:

Route A	Route B
10 minutes	12 minutes
Poor lighting	Better lighting
More reported incidents	Fewer reported incidents
	Safe places nearby



A normal navigation app may prefer Route A because it is faster.

SafeRoute allows the user to see this safety difference and choose accordingly.

Therefore:

SafeRoute adds safety information to normal navigation.

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SCOPE

What Can SafeRoute Do?

1 **Navigation**

Find and compare different routes.

2 **Safety Analysis**

Give every route a Safety Score from 1–10.

8.4
Safety Score

3 **Incident Reports**

Users can report unsafe conditions such as:

- Poor lighting
- Snatching
- Harassment
- Suspicious activity
- Isolated roads

4 **Safe Havens**

Find nearby:

Police Stations | Hospitals | Pharmacies | Petrol Pump | Fire Station

5 **Community Ratings**

Allow users to rate the safety of roads and locations.

6 **SOS**

Send an emergency alert with the user's location.

7 **Map Views**

Standard | Satellite | Dark | Terrain | Transport

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OBJECTIVES

Objectives of SafeRoute

1. Help users find safer routes.
2. Compare the fastest and safest routes.
3. Give each route a clear Safety Score.
4. Show reported safety problems.
5. Help users find nearby safe places.
6. Allow users to rate the safety of roads and locations.
7. Provide quick help through the SOS button.
8. Make navigation more safety-focused and informative.



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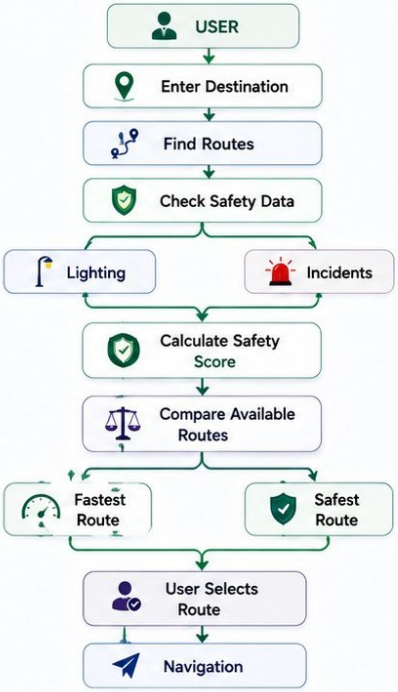
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SYSTEM DESIGN

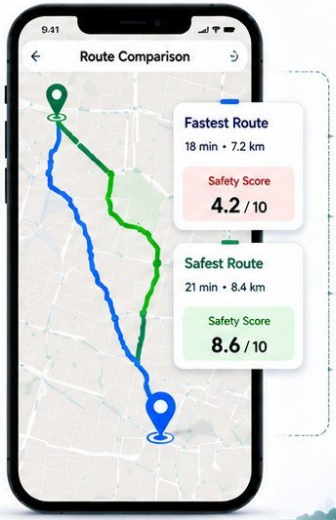
How Does SafeRoute Work?



SafeRoute checks multiple safety factors

Street Lighting	Checks availability of street lighting on the route
Reported Incidents	Considers number of incidents reported in area
Community Ratings	Uses user ratings for roads and locations
Safe Places Nearby	Checks distance from hospitals, police stations, etc.





- SafeRoute evaluates routes using real safety data
- Shows both fastest and safest options
- Empowers user to make the right choice
- Always ready to help in emergencies

Simple Explanation

SafeRoute finds routes → checks safety → calculates scores → compares routes → helps the user choose.

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TECHNOLOGIES

Technologies Used

Frontend

- **React.js** – Builds the website interface
- **Vite** – Fast development and building
- **Tailwind CSS** – Creates the design
- **Framer Motion** – Adds animations
- **Leaflet** – Displays interactive maps

Backend

- Node.js
- Express.js
- Socket.io

Used for handling requests and real-time location updates.

Database

- MongoDB
- Mongoose

Stores:

- Users
- Incidents
- Emergency contacts
- Other application data

APIs

- **OpenStreetMap** – Map data
- **OSRM** – Route generation
- **Nominatim** – Location/address search
- **Overpass API** – Safe-haven locations

Deployment

Render



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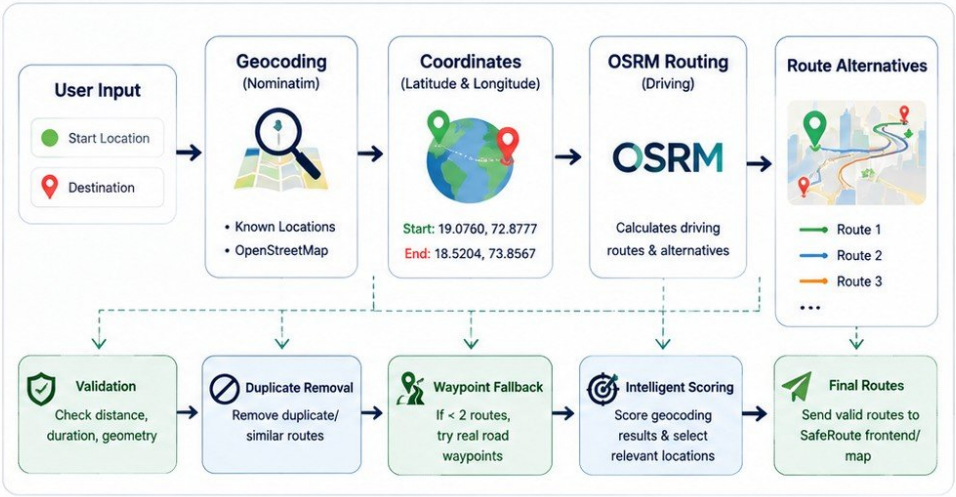
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Backend Routing & Geocoding

OSRM + Nominatim Routing Service

- Converts user-entered locations into **latitude & longitude** using known locations or **Nominatim/OpenStreetMap**.
- Uses **OSRM** to calculate driving routes and alternative routes.
- Validates route **distance**, **duration**, and **geometry**.
- Detects and removes **duplicate/similar routes**.
- If fewer than 2 genuine routes are found, it tries **real road waypoints** to find additional alternatives.
- Selects relevant locations using intelligent **geocoding result scoring**.
- Sends the final valid routes to the **SafeRoute frontend/map.**



FLOW

User Input → Geocoding → Coordinates → OSRM → Route Alternatives → Validation & Duplicate Removal → Final Routes → SafeRoute Map

Presentation line:

"This module is the routing engine of SafeRoute. It converts **locations** into **coordinates**, finds **genuine driving alternatives**, **removes duplicate routes**, and sends the best available routes to the map."

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SafeRoute – Safety Scoring Algorithm

Weighted Sum Model (WSM)

Purpose: Evaluate and rank OSRM-generated routes based on multiple safety factors.

OSRM Route Alternatives

↓

Calculate Safety Factors

Factor	Weight	Evaluation
Community Incidents	40%	Active incidents within 250 m; severity, type, confidence, recency & confirmations
Lighting / Road Conditions	30%	Road characteristics + additional night-time penalty
Time of Day	20%	Safety varies according to travel hour
Road Classification	10%	Primary, secondary, residential, service roads, paths, etc.

EXAMPLE: ROUTE A

Safety Factor	Score (0 – 1)	Weight	Weighted Score
Community Incidents	0.90	× 40%	0.360
Lighting	0.80	× 30%	0.240
Time of Day	1.00	× 20%	0.200
Road Type	0.85	× 10%	0.085

Calculation: (0.90 × 0.40) + (0.80 × 0.30) + (1.00 × 0.20) + (0.85 × 0.10)

= 0.885

Final Safety Score = 0.885 × 10 = 8.9 / 10

↪ **Route A = Very Safe**

ROUTE SELECTION PROCESS

OSRM Alternatives → Safety Scoring → Weighted Sum → Score (0 – 10) → Compare Routes → Safest Route

SCORE INTERPRETATION

8.5 – 10	7 – 8.4	5.5 – 6.9	4 – 5.4	< 4
Very Safe	Safe	Moderate	Use Caution	High Risk

POINT

Higher score = safer route.

SafeRoute combines safety factors rather than selecting a route only by distance or travel time.



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UML DIAGRAMS

System UML Diagrams

Use Case Diagram
Shows what users can do in SafeRoute.

Activity Diagram
Shows the step-by-step working of the system.

Sequence Diagram
Shows how different parts of the system communicate.

Class Diagram
Shows the main objects and their relationships.

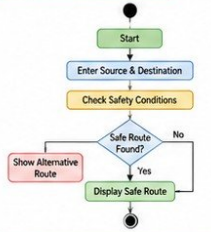
Component Diagram
Shows the major parts of the SafeRoute software.

Deployment Diagram
Shows where the frontend, backend, database and APIs are deployed.

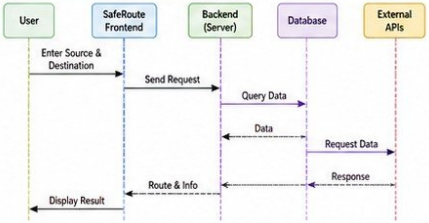
1 Use Case Diagram



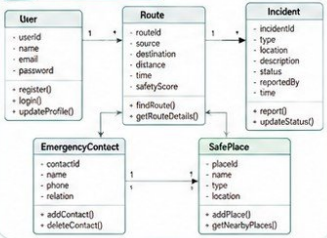
2 Activity Diagram



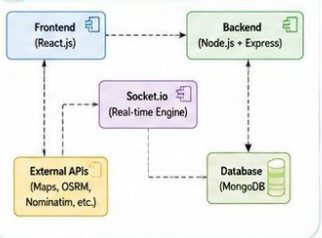
3 Sequence Diagram



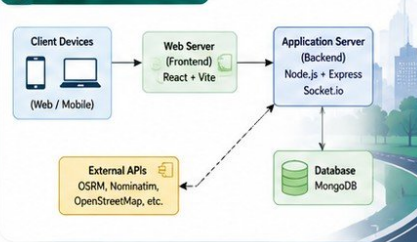
4 Class Diagram



5 Component Diagram



6 Deployment Diagram



Main System Flow

User → SafeRoute → Backend → Database / APIs → SafeRoute → User

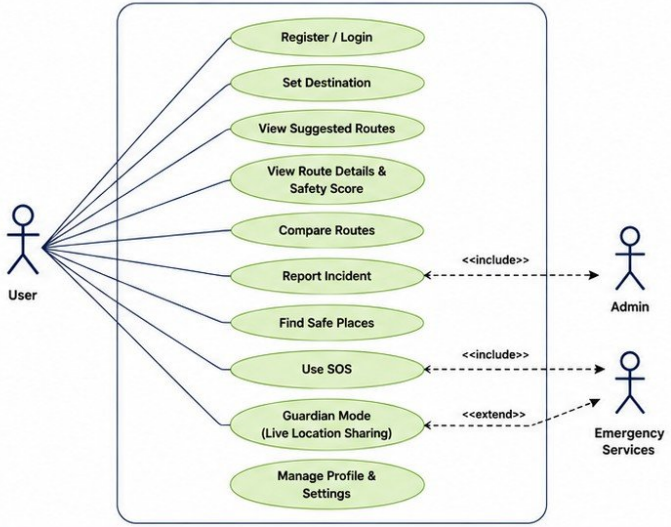
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UML DIAGRAMS

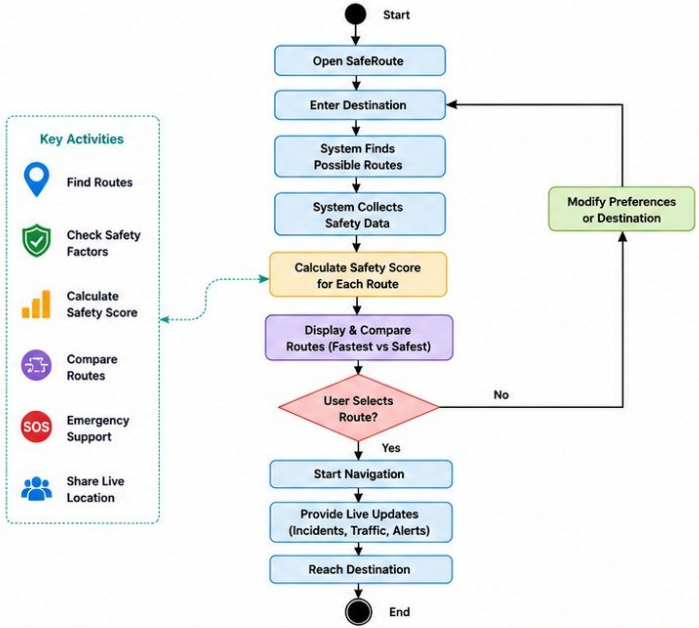
1 USE CASE DIAGRAM

Shows what users can do in SafeRoute.



2 ACTIVITY DIAGRAM

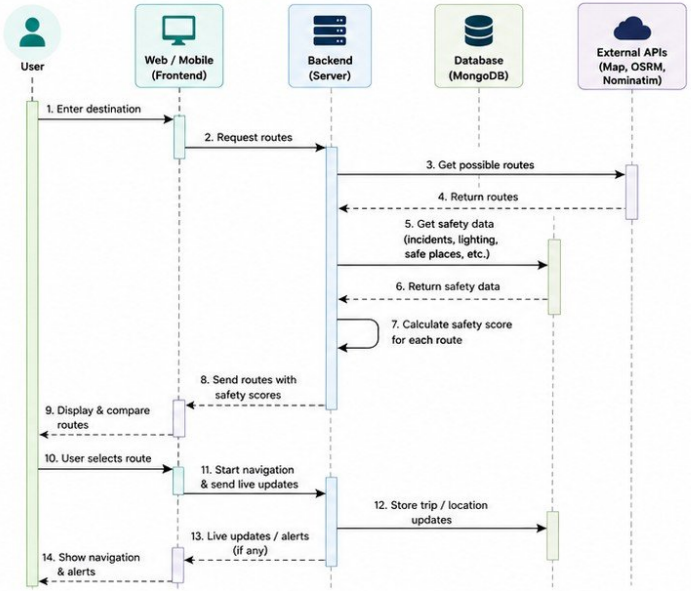
Shows the step-by-step working of the system.



UML DIAGRAMS

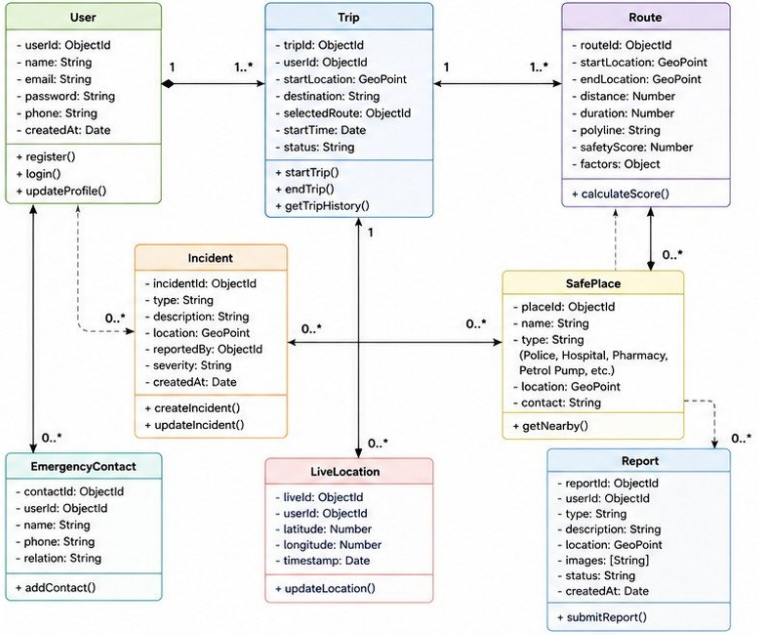
3 SEQUENCE DIAGRAM

Shows how different parts of the system communicate.



4 CLASS DIAGRAM

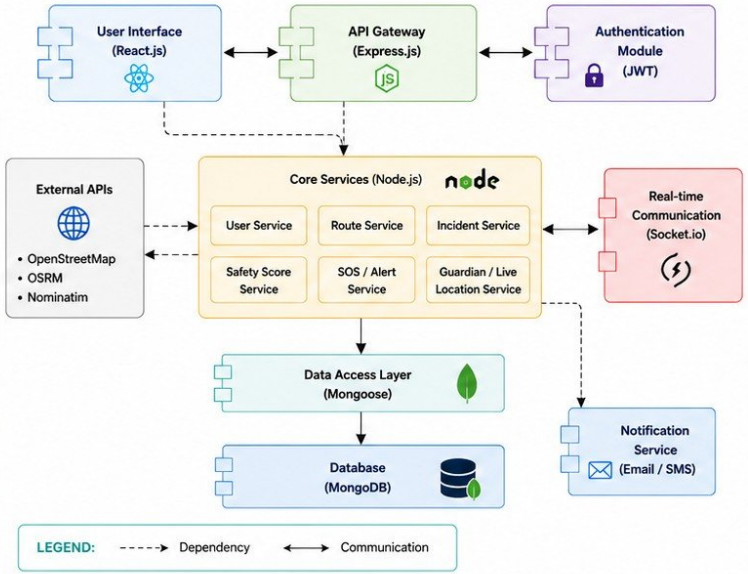
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UML DIAGRAMS

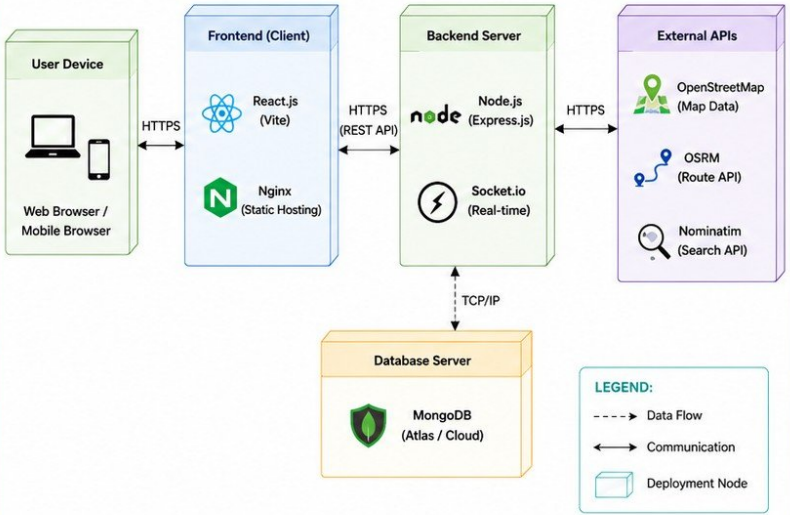
5 COMPONENT DIAGRAM

Shows the major parts of the SafeRoute software.



6 DEPLOYMENT DIAGRAM

Shows where the frontend, backend, database and APIs are deployed.



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FUTURE SCOPE

Future Improvements

1 AI Safety Prediction
Predict dangerous areas using previous safety data.

2 Real-Time Information
Use live crime, traffic and environmental information.

3 Mobile App
Create dedicated Android and iOS applications.

4 Voice Navigation
Allow users to navigate without looking at the screen.

5 Automatic Emergency Alerts
Automatically alert contacts when unusual situations are detected.

6 Guardian Mode
Allow to Share live location with selected contacts.

7 Personalized Routes
Suggest routes based on the user's safety preferences.

8 More Locations
Expand SafeRoute to more cities and regions.



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CONCLUSION

SafeRoute improves traditional navigation by adding safety information.

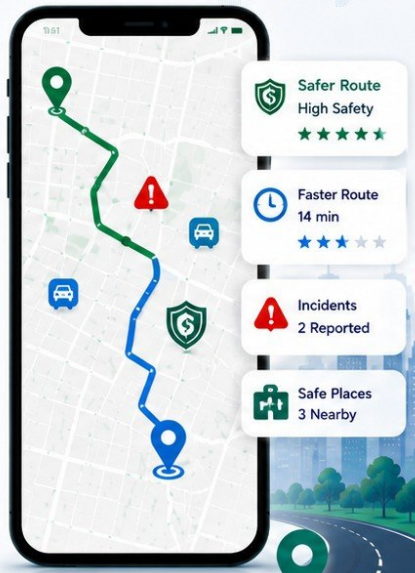
It helps users:

Find routes	Find the best routes to any destination.
Understand route safety	Know how safe each route is before you travel.
Compare routes	Compare multiple routes based on time and safety.
Report incidents	Report accidents, crimes or other unsafe situations.
Find safe places	Locate nearby hospitals, police stations and safe places.
users to rate the safety of roads and locations.	Help others by sharing your experience and ratings.
SOS Get emergency assistance	Quick access to emergency help and contacts.

The Main Idea

> "The shortest path isn't always the safest."

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THANK YOU

Any Questions?

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Together, let's navigate smart and travel safe!